Marginal or Average: Which Metric Really Matters?

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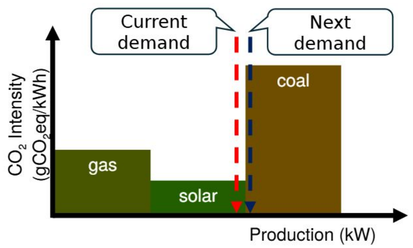
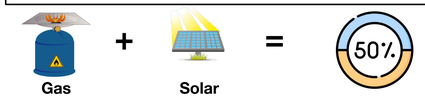
Background: There are two types of carbon intensity signals: average and marginal. Both signals provide different information about the electricity grid operations. Unfortunately, there is a lack of consensus on the "right" signal for carbon-aware optimizations. As a result, we conduct a data analysis to understand the implications of signal selection on carbon-aware optimization techniques and how we might be able to reduce carbon emissions.

Two Common Ways of Accounting Carbon Intensity



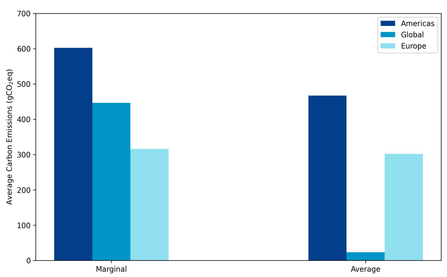
***Average Carbon Intensity**: Weighted average of carbon emissions and their productions*



***Marginal Carbon Intensity**: Carbon intensity with respect to the change in demand*

Since the next demand is served by coal, the marginal carbon intensity is that of coal

The BIG Picture: Average and Marginal Carbon Intensity

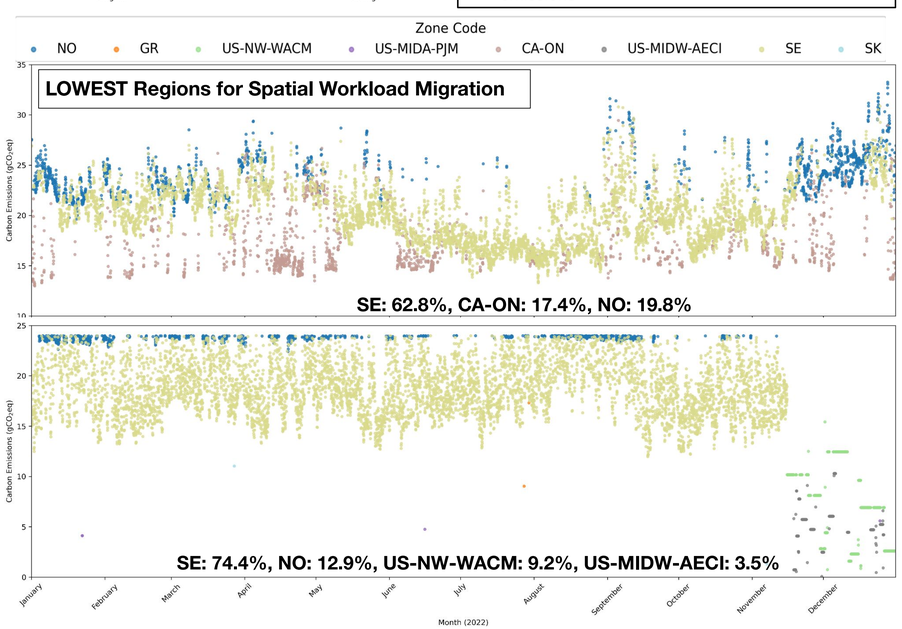


The mean carbon emissions of the two carbon intensity signals among three different scales:

America: 99 regions
Global: 3 regions
Europe: 31 regions

Takeaways:

1. Marginal carbon emissions are higher than average across all three scales
2. At a global scale, the average carbon emissions is significantly lower than the marginal carbon emissions



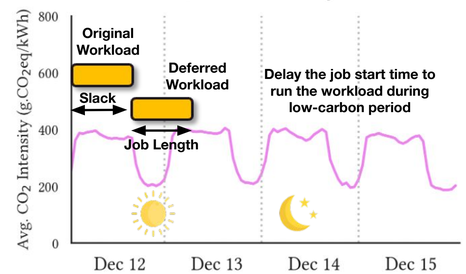
Regions to migrate the workload for the maximum carbon at each time period in the year 2022

Takeaways:

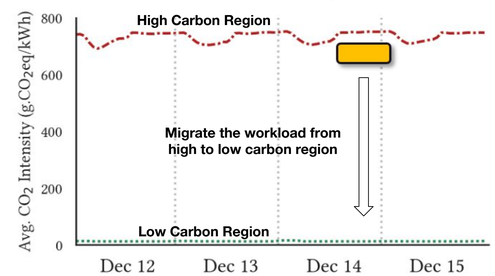
1. Sweden is the lowest region across two signals
2. Apart from Sweden, the two signals have different choice of destination regions

Spatiotemporal Workload Shifting

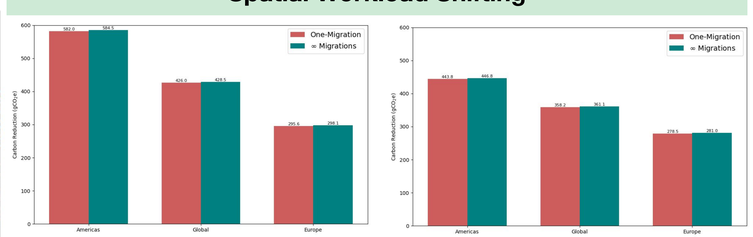
Temporal Shifting



Spatial Shifting



Spatial Workload Shifting



Marginal carbon intensity as a scheduling signal

Average carbon intensity as a scheduling signal

Determine whether sophisticated migration techniques are necessary for spatial migration

Migration Techniques:
One migration:

- Migrate the workload once to the region with the lowest mean carbon intensity: Sweden (SE) for both average and marginal carbon intensity

Infinite migration:

- Migrate the workload to the lowest region at each time period for the whole year

For each technique, the destination region is determined by either the average or the marginal carbon intensity signal

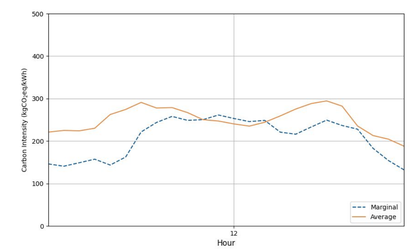
Takeaways:

1. The majority of the carbon reduction comes from migrating once
2. Marginal carbon intensity has higher carbon reductions across all three scales

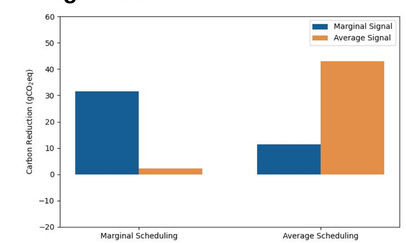
Temporal Workload Shifting

- Choose one signal as a scheduling signal for a deferrable workload with 2 hours job length and 4 hours slack.
- Evaluate the carbon reduction from both the scheduling and non-scheduling signals' perspective

Austria: Interchange in Magnitude Order

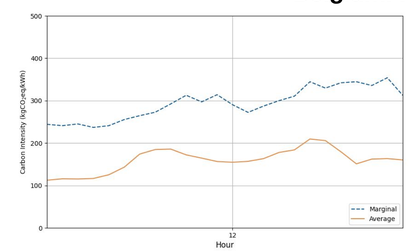


Carbon intensities for Austria

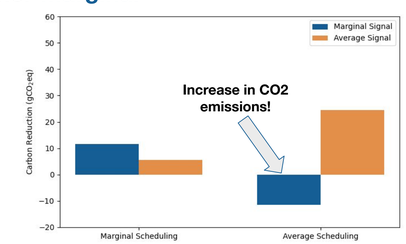


Carbon reduction for Austria

Belgium: Higher Marginal

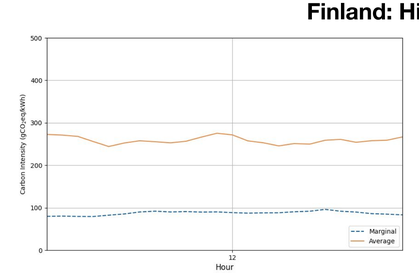


Carbon intensities for Belgium

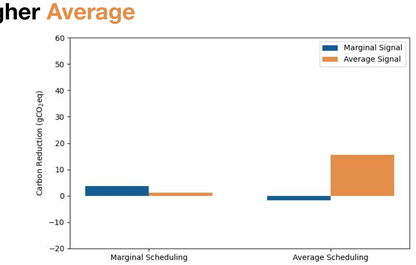


Carbon reduction for Belgium

Finland: Higher Average



Carbon intensities for Finland



Carbon reduction for Finland

Takeaways:

1. Regardless of the signal characteristics, the scheduling signal will always results in higher carbon reduction as compared to the other signal's perspective
2. There could be an increase in carbon emissions based on the non-scheduling signal's perspective
3. Regions with high variations enjoy the high carbon reductions

Conclusion

- Our research demonstrates the differences between two commonly used carbon signals, marginal and average intensities
- The two signals yield distinct outcomes when they are used as the scheduling signals for temporal and spatial workload shifting
- Optimizing for one signal can result in higher carbon emissions from the other signal's perspective